Predicting the full-load power output of a combined cycle power plant

Machine learning regression on 9,527 hours of plant data

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Scan for the live Tableau dashboard

3.45 MW
test RMSE of the selected model (KNN)

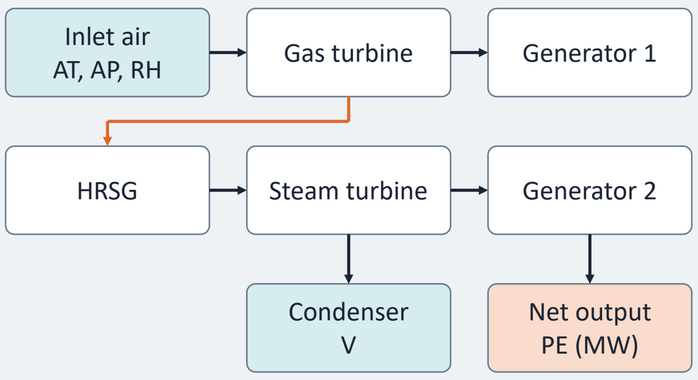
0.540%
test MAPE, against 0.872% in the reference study

0.959
test R^2, against 0.912 in the reference study

9,527
hours of full-load plant data after cleaning

1 The problem

At full load the plant's net output still moves between 420 and 496 MW with the weather. We predict it from four hourly measurements: a supervised regression task.



Teal: measured inputs. Orange: hot exhaust.

- Grid commitment: the plant promises its MW in advance, and over-prediction is penalized.
- Performance monitoring: output below a healthy-plant prediction signals fouling or wear.

2 Data and preprocessing

41 duplicate records dropped, so no hour is in both train and test sets

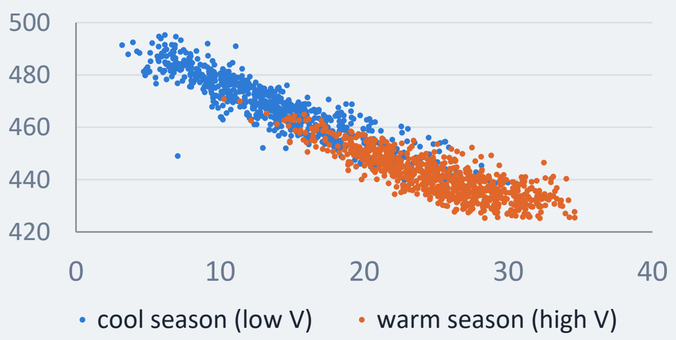
55 humidity readings above 100% capped, rows kept

182 hours with V above air pressure: the unit label is wrong, V kept as an index

0 outliers deleted: they are real weather

Source: UCI Combined Cycle Power Plant dataset, 9,568 hours, 2006 to 2011. Split 80/20 stratified on output; scaler inside the pipeline; tuning by 5-fold cross-validation on training hours; test set used once.

3 Engineering insights



- −2.17 MW per °C; the hottest hours give 52 MW less
- Humid air lowers output at a fixed temperature
- The condenser penalty falls mainly at 15 to 30 °C

4 Three nonlinear models

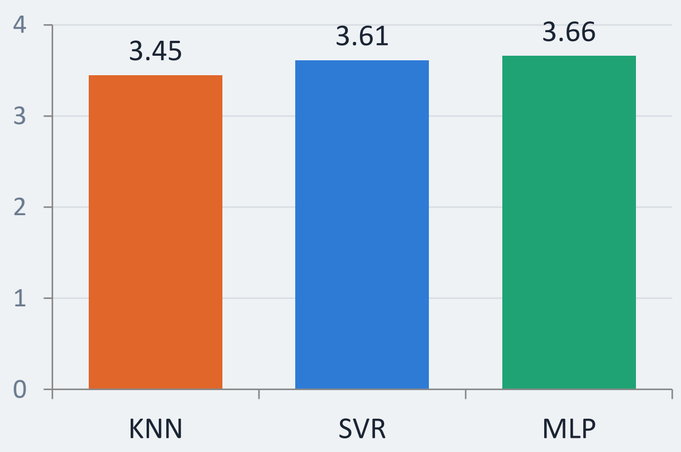
None was used in the reference study. All were tuned by 5-fold cross-validation on the 7,621 training hours.

SVR — **Support vector regression**
RBF kernel, C = 30, γ = 2, ε = 2 MW; CV RMSE 3.83 MW

KNN — **k-nearest neighbours**
k = 6, distance-weighted, Manhattan; CV RMSE 3.64 MW

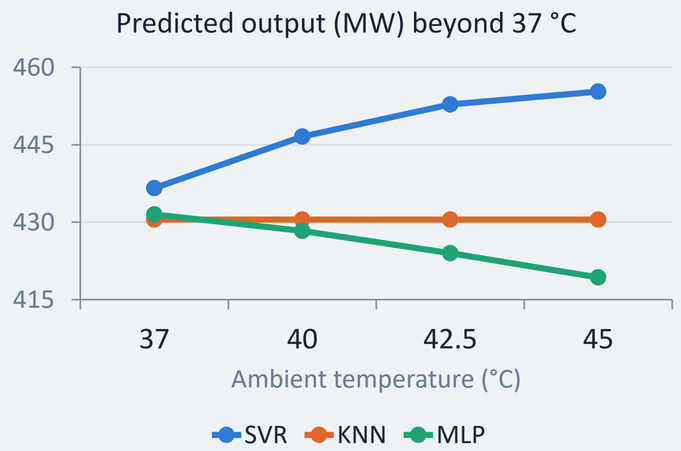
MLP — **Neural network**
128-128-64 ReLU, early stopping; CV RMSE 3.80 MW

5 Results on unseen hours



- All three beat the reference study (R^2 0.912, MAPE 0.872%)
- KNN's lead is significant in a paired bootstrap
- KNN over-predicts by more than 5 MW on the fewest hours: 5.3%, against 6.9% (SVR) and 7.8% (MLP)

6 Accuracy is not enough



Physics says output keeps falling with heat. The SVR predicts the opposite: 455 MW at 45 °C. No model is trusted outside its data, so the deployed model flags such hours.

7 Selected model: KNN

- Fewest costly over-predictions
- Most accurate on unseen hours
- Each prediction rests on six real past hours
- Range and density checks flag hours it cannot judge

Five new operating conditions

Case	KNN (MW)	Status
Hot afternoon, 33 °C	433.9	valid
Cold night, 6 °C	487.9	valid
Mild humid day	453.3	valid
Mild dry day	452.6	caution
Heatwave, 41 °C	429.3	flagged

8 What the model learned

−54 MW AT across its range

−10 MW V, mostly at 15 to 30 °C

−7 MW RH, dry to humid

≈ 3 MW AP, no clear direction

Accumulated local effects of the MLP match thermodynamics.

9 Two plant decisions

- Alarm when output stays more than 5.2 MW below the prediction
- Commit 2.4 MW below the prediction when a shortfall costs four times the sale price

Conclusion

Four plant measurements predict full-load output to 3.45 MW. The model beats the reference study, behaves as physics predicts, and drives an alarm and a grid safety margin.

Data: Tüfekci, P. & Kaya, H., Combined Cycle Power Plant, UCI Machine Learning Repository, CC BY 4.0, accessed 24 October 2026. Reference study: Santarisi, N. S. & Faouri, S. S. (2021), Eastern-European Journal of Enterprise Technologies, 6(8(114)), 16-26. Tools: Python, scikit-learn, Tableau Public.